

LONDON HEATHROW, United Kingdom

WMO: 037720

Lat: 51.4792N

Lon: 0.4506W

Elev: 83

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	30.0	21.2	15.9	32.9	23.7	17.9	33.9	29.4	50.5	26.4	49.6	5.7	270	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	83.7	65.7	79.7	64.1	76.4	62.8	67.6	79.3	65.9	75.8	64.4	73.2	9.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	88.0	70.9	62.1	83.6	69.4	60.8	79.9	68.1	32.0	79.3	30.6	76.0	29.5	73.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.6	18.3	DB	23.3	89.9	2.5	4.0	21.5	92.8	20.0	95.1	18.6	97.4	16.8	100.3	(4)
(5)			WB	22.4	70.6	2.5	1.5	20.6	71.6	19.2	72.5	17.8	73.3	16.0	74.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	41.9	42.7	46.0	50.5	56.2	61.9	66.0	65.3	60.4	54.3	47.2	42.9
	DBStd	10.11	5.69	5.63	5.32	5.48	5.52	4.93	4.97	4.65	4.34	4.99	5.53	6.34
	HDD50	1040	254	210	143	58	9	0	0	0	0	18	116	232
	HDD65	4562	715	625	589	434	279	119	47	51	150	333	534	686
	CDD50	2136	4	5	19	74	201	356	497	475	311	151	32	11
	CDD65	183	0	0	0	0	5	26	79	61	11	1	0	0
	CDH74	1290	0	0	0	8	50	215	563	387	62	5	0	0
	CDH80	311	0	0	0	0	3	48	148	104	7	1	0	0
Wind	WSAvg	8.8	9.9	9.6	9.3	8.6	8.7	8.7	8.5	8.2	8.1	8.7	8.4	9.4
Precipitation	PrecAvg	23.7	2.1	1.4	1.7	1.9	1.9	2.0	1.7	2.0	2.1	2.4	2.3	2.2
	PrecMax	32.3	4.5	3.6	3.7	4.0	4.0	5.0	3.7	5.9	5.5	6.9	5.9	4.7
	PrecMin	16.6	0.4	0.1	0.2	0.2	0.2	0.2	0.3	0.0	0.1	0.1	0.3	0.4
	PrecStd	3.9	1.1	0.8	1.0	0.9	1.0	1.2	0.9	1.2	1.5	1.5	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	58.7	64.5	74.3	78.9	85.8	88.8	88.2	80.1	70.9	61.6	57.9
		MCWB	52.8	51.7	52.7	58.2	62.3	66.1	67.8	67.8	64.4	62.3	57.9	54.6
	2%	DB	54.3	54.8	60.0	67.6	74.8	79.8	84.1	82.7	75.0	66.3	59.2	56.0
		MCWB	51.4	50.3	51.1	55.3	60.9	63.7	65.6	66.2	62.3	59.6	55.8	53.2
	5%	DB	52.4	52.8	56.9	63.4	70.6	75.5	80.1	78.1	71.1	63.8	57.4	54.5
		MCWB	49.6	48.9	50.1	53.1	58.8	61.8	64.2	64.0	60.7	58.1	54.0	51.9
	10%	DB	50.6	51.0	54.4	60.0	66.4	72.0	76.4	74.3	68.3	61.8	55.6	52.5
		MCWB	47.9	47.4	48.9	51.3	56.3	60.1	62.9	62.6	59.4	57.1	52.4	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	53.1	54.7	59.6	65.2	68.3	69.8	69.7	66.9	63.2	58.8	55.5
		MCDB	55.4	56.2	60.4	71.0	75.4	81.4	85.3	84.1	75.2	69.4	60.9	56.9
	2%	WB	51.9	51.2	52.7	56.5	62.0	65.4	67.5	67.5	64.5	61.0	56.5	53.9
		MCDB	53.9	54.0	57.5	65.5	71.1	76.0	79.7	78.9	72.1	64.6	58.7	55.7
	5%	WB	50.0	49.5	51.3	54.1	59.8	63.2	65.7	65.7	62.6	59.4	54.6	52.1
		MCDB	52.1	52.1	55.5	61.0	68.5	72.3	76.3	74.7	68.2	62.7	57.0	54.2
	10%	WB	48.3	48.0	49.8	52.3	57.7	61.5	64.1	64.0	61.1	58.0	52.8	50.1
		MCDB	50.4	50.6	53.7	58.6	65.1	70.2	74.0	71.6	66.2	61.2	55.2	52.4
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	12.6	14.9	15.5	16.0	16.3	15.6	14.4	12.0	10.0	8.9
		MCDBR	10.3	11.8	16.3	20.1	21.5	22.1	22.5	21.1	18.3	13.5	10.1	8.8
		MCWBR	9.7	8.7	9.8	10.3	10.4	10.0	8.5	8.2	8.4	8.2	8.2	8.2
	5% WB	MCDBR	10.1	10.7	14.2	18.0	19.3	19.2	19.4	18.1	15.3	12.0	9.9	9.0
		MCWBR	9.8	9.0	9.6	10.1	10.1	9.8	8.5	8.3	8.7	8.6	8.8	8.8
Clear-Sky Solar Irradiance	taub	0.332	0.347	0.384	0.406	0.410	0.410	0.405	0.396	0.378	0.361	0.335	0.320	
	taud	2.364	2.369	2.260	2.220	2.250	2.276	2.297	2.341	2.387	2.415	2.409	2.350	
	Ebn at Noon	213	244	254	263	268	268	267	264	255	236	211	196	
	Edh at Noon	20	26	35	40	41	40	39	36	30	24	19	17	
All-Sky Solar Radiation	RadAvg	259	466	810	1250	1515	1677	1588	1320	996	578	308	203	
	RadStd	22	62	111	145	97	170	156	108	85	52	35	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page